

Gogal Vijay B

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PROFESSIONAL SUMMARY

Software Engineer (M.Tech, IISc) with hands-on experience in **backend development**, **systems programming**, and **distributed systems**. Built production payment infrastructure at Juspay and independently engineered low-level systems spanning OS internals, network security, and storage engines. Comfortable working across backend services, high-performance C/C++ systems, and Linux-level infrastructure.

TECHNICAL SKILLS

- **Languages:** C++ (11/14/17), C, Python, SQL, JavaScript, Java.
- **Backend and Distributed Systems:** REST APIs, System Design, Distributed Systems, Microservices, API Design.
- **Databases and Storage:** PostgreSQL / MySQL, Redis, LSM Trees, WAL, Indexing
- **Systems Programming:** Linux Internals, Concurrency, Multithreading, Socket Programming, epoll
- **Tools and CS Fundamentals:** Git, Docker, GDB, Linux, Data Structures & Algorithms, Operating Systems, Computer Networks

WORK EXPERIENCE

Juspay Technologies

Associate Software Development Engineer

Bengaluru, India

Jan 2025 – Nov 2025

- **Payment Infrastructure (EMI Team):**
 - Owned and delivered **two complete EMI payment integrations end-to-end** — designing request/response contracts, modifying **PostgreSQL** schemas, handling **error response mapping** across transaction lifecycle states, and working with **Redis** for caching through to production readiness independently.
 - Integrated **API endpoints** handling the full EMI transaction flow end-to-end, implemented **webhook handlers** for real-time provider callbacks, and added **retry logic** for transient failures with structured error responses across failure states.
- **AI Team (Internal Developer Tooling):**
 - Contributed **backend API features and frontend automation** to an internal LLM-based developer tool adopted across Juspay's **engineering org of 2000+** as the primary AI-assisted workflow tool.

PROJECTS

TeleportVM — Distributed Shared Memory Engine | C, Linux, Systems Programming, Distributed Systems [GitHub]

- Built a userspace **distributed shared memory engine** in C using Linux's **userfaultfd** API to transparently unify virtual address spaces across networked machines, resolving missing pages over TCP without any special application-level networking code.
- Implemented a **coordinator-based MSI cache coherence protocol** with per-page locking, non-blocking **epoll**-driven network engine, custom binary wire protocol, and clock-algorithm eviction with writeback — verified via a scripted write-write-read conflict harness asserting zero staleness across nodes.

Custom VPN Daemon | C, Systems Programming, Network Security

[GitHub]

- Built a **VPN daemon in C** using **Linux TUN** interfaces and **UDP transport** with **X25519 Diffie-Hellman** key exchange, **ChaCha20-Poly1305 AEAD** encryption via libsodium, and **64-bit sliding window** replay protection.
- Engineered **concurrent I/O** via **epoll** and **pthread**s handling multiple peers on a single thread — with **ping/pong heartbeat**, **routing table hijacking**, **DNS leak prevention**, and full daemonization with syslog integration.

Distributed Log-Structured Merge (LSM) Tree | C++, Systems Programming, Storage Engine

[GitHub]

- Engineered a **production-grade key-value database engine** in C++ using **LSM Tree** architecture — the same storage model underlying **RocksDB**, **LevelDB**, and **Cassandra** — with disk-backed persistence and crash recovery via **WAL**.
- Implemented **MemTable**, **SSTables**, **Bloom filters**, **sparse indexing**, and **multi-level compaction**, reducing **disk reads by 30–40%** during point lookups and lowering read latency under mixed workloads.

EDUCATION

Indian Institute of Science (IISc), Bengaluru

MTech, Computer Science and Engineering

Aug 2022 – Dec 2024

AIR 274 in GATE 2022

Kamaraj College of Engineering and Technology, Virudhunagar

BE, Computer Science and Engineering

Aug 2016 – Apr 2020